Routing, capacity and width in dynamic convolution: ConvNeXt-Tiny

Replication of the 132-run ResNet-18 program

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Generated from result files by `report/build_model_report.py`

SYNTHETIC TEST DATA. Every number in this document is generated to test the pipeline. None is a result.

This document reports one backbone of the cross-architecture replication. It is generated from the result files: every number, table and figure is computed, and every sentence states what a test returned. Interpretation across backbones is in the cross-model report. Method, statistical protocol and the pre-registered margin are those of the ResNet-18 study and are not repeated here.

1 Setup and coverage

Backbone. ConvNeXt-Tiny, adapted to 32×32 and 64×64 inputs as recorded in `src/models`. Width 64 is the standard network; widths 32 and 16 scale every channel count by $1/2$ and $1/4$. Dynamic layers follow the ResNet-18 rule literally (every $k \times k$ convolution with $k > 1$ inside the blocks; stem, 1×1 convolutions, shortcuts, downsampling layers and head static): here the 7×7 depthwise convolution of every block, 18 layers. Initialisation: truncated normal, std 0.02, for every static convolution and every bank kernel.

Size. At CIFAR-100 width 64 the static network stores 27.89 M parameters and the $K = 4$ `gap` network 29.70 M, of which 817.1 K are router parameters (Table 14). The bank adds 0.99 M to the static network; the router is 2.9 % of the static network's size. The router is the ResNet-18 design (a two-layer MLP with hidden width $C_{\text{in}}/4$ on the global descriptor), so its size grows with the square of a layer's input channels while a depthwise kernel grows linearly: where dynamic layers are depthwise, the routed arms carry many more parameters than the bank. The `gap−uniform` routing term includes them.

Gate. The first run, static, CIFAR-100, width 64, seed 0, reached 68.36 % against a pre-registered floor of 60.0 %: passed.

Coverage. 130 of 132 runs are present; 2 are missing, of which 1 diverged (non-finite training loss). 0 group(s) ran with gradient accumulation because batch 128 did not fit in memory. Table 1 lists every missing run and every deviation. Paired tests use the seeds both arms have; a verdict computed on fewer than six pairs carries its n.

Table 1: Missing runs, divergences and deviations from the recipe.

dataset	width	K	τ	arm	seed	status
CIFAR-100	32	4	const	`gap`	1	diverged (non-finite loss)
Tiny-ImageNet	64	4	const	`static`	5	not run

2 CIFAR-100 at the standard width

Against `static`: `uniform` +0.066 pp (equivalent); `gap` +0.736 pp (higher); `shuffled` -0.152 pp (equivalent); `frozen_random` +0.221 pp (inconclusive). 2 of the four arms are equivalent to `static` within 0.5 pp. The largest gap between arm means is 0.888 pp against a median within-arm seed standard deviation of 0.259 pp; one-way ANOVA across the five arms $F = 10.329$, $p = 0.000$. `frozen_random−gap` $= -0.515$ pp ($p = 0.013$, $p_{\text{TOST}} = 0.5424$). Power for `gap−uniform` here: paired sd 0.346 pp, minimum detectable difference 0.363 pp, simulated power 30.2, 80.7 and 100.0 % at true effects of 0.25, 0.5 and 1.0 pp. Table 2 gives the routing entropy by arm.

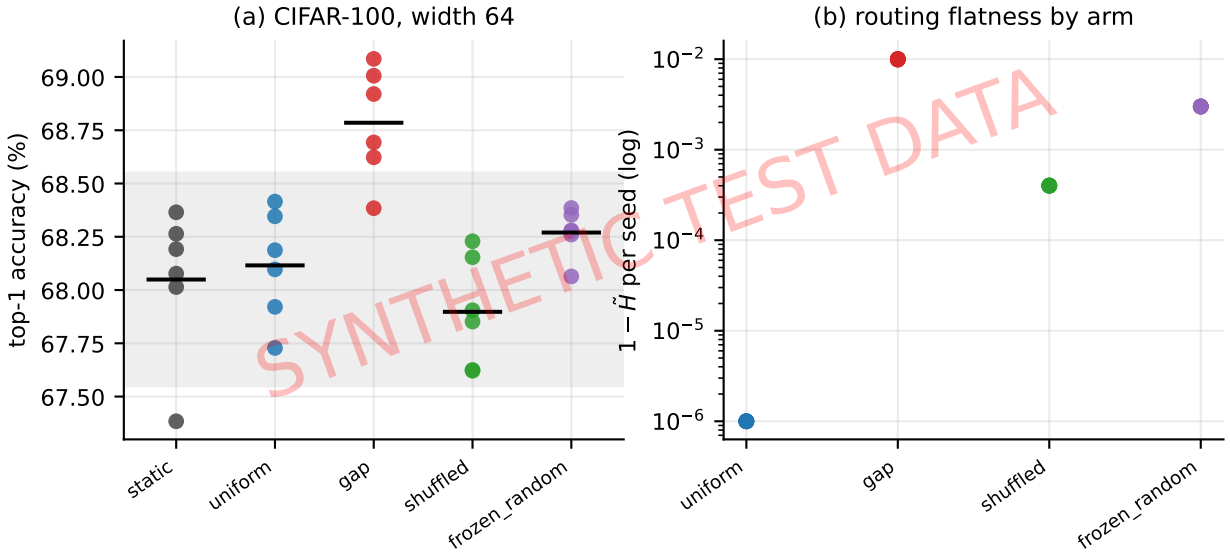


Figure 1: CIFAR-100, width 64. (a) Every run of the five arms; bars are arm means and the shaded band is ±0.5 pp around `static`. (b) Router flatness $1 - \tilde{H}$ per seed, log scale; `uniform` is exactly zero and is drawn at the axis floor.

Table 2: Routing entropy by arm, CIFAR-100 width 64, per-seed means.

arm	mean $\tilde{H}$	min	max	$1 - \tilde{H}$
`uniform`	1.0000	1.0000	1.0000	0.00000
`gap`	0.9900	0.9899	0.9901	0.00998
`shuffled`	0.9996	0.9996	0.9996	0.00040
`frozen_random`	0.9970	0.9970	0.9970	0.00300

3 The capacity sweep on CIFAR-100

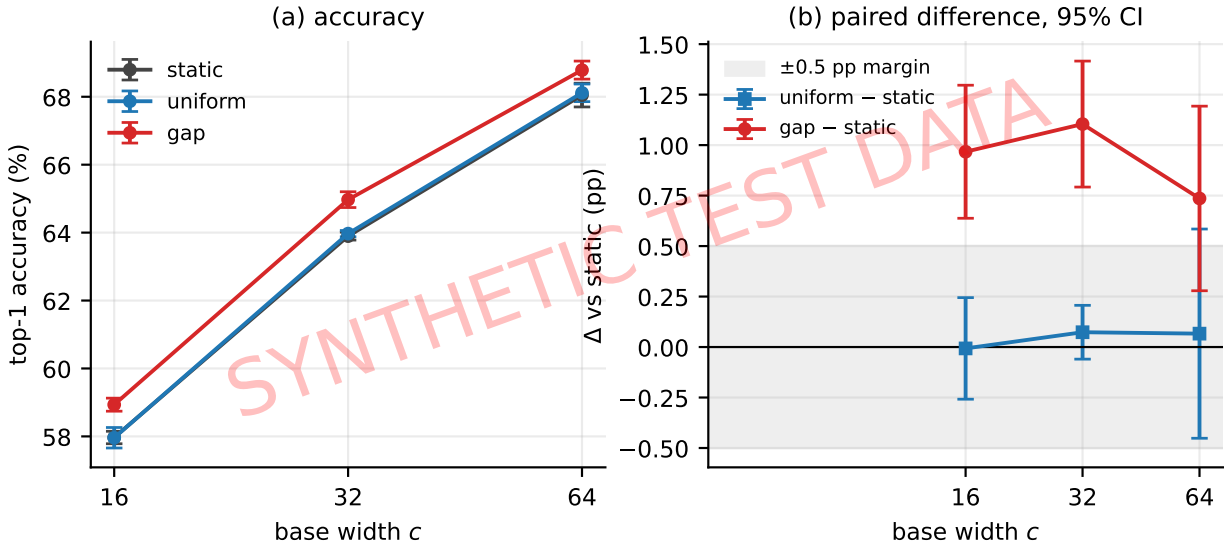


Figure 2: CIFAR-100. (a) Accuracy against base width, ±1 standard deviation over six seeds. (b) Paired difference against `static` with 95 % confidence intervals; the band is the ±0.5 pp margin.

`gap−static`: +0.967 pp at width 16 (higher), +1.104 pp at width 32 (higher (n=5)), +0.736 pp at width 64 (higher) (Figure 2, Table 13).

4 Bank against routing

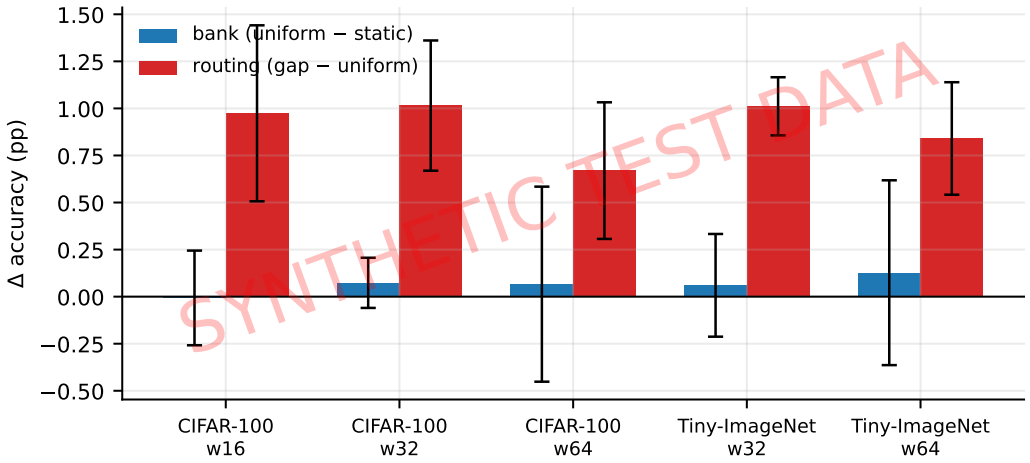


Figure 3: The total effect split into a bank term (`uniform−static`) and a routing term (`gap−uniform`), five conditions, 95 % confidence intervals.

The bank term is equivalent to zero in 4 of 5 conditions; the others: Tiny-ImageNet width 64: inconclusive (n=5) (+0.127 pp). The routing term is positive with $p < 0.05$ in 5 of 5 conditions: CIFAR-100 width 16 +0.974 pp ($p = 0.0030$); CIFAR-100 width 32 +1.015 pp ($p = 0.0012$); CIFAR-100 width 64 +0.669 pp ($p = 0.0051$); Tiny-ImageNet width 32 +1.011 pp ($p = 0.0000$); Tiny-ImageNet width 64 +0.840 pp ($p = 0.0008$) (Table 3, Figure 3).

Table 3: Decomposition. Bank verdicts use TOST at ±0.5 pp.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	-0.007	0.0020	equivalent	+0.974	0.0030	+0.967	0.0006
CIFAR-100	32	+0.073	0.0002	equivalent	+1.015	0.0012	+1.104	0.0006
CIFAR-100	64	+0.066	0.0420	equivalent	+0.669	0.0051	+0.736	0.0090
Tiny-ImageNet	32	+0.060	0.0045	equivalent	+1.011	$< 10^{-4}$	+1.071	$< 10^{-4}$
Tiny-ImageNet	64	+0.127	0.0513	inconclusive (n=5)	+0.840	0.0008	+0.953	0.0033

5 Bank size

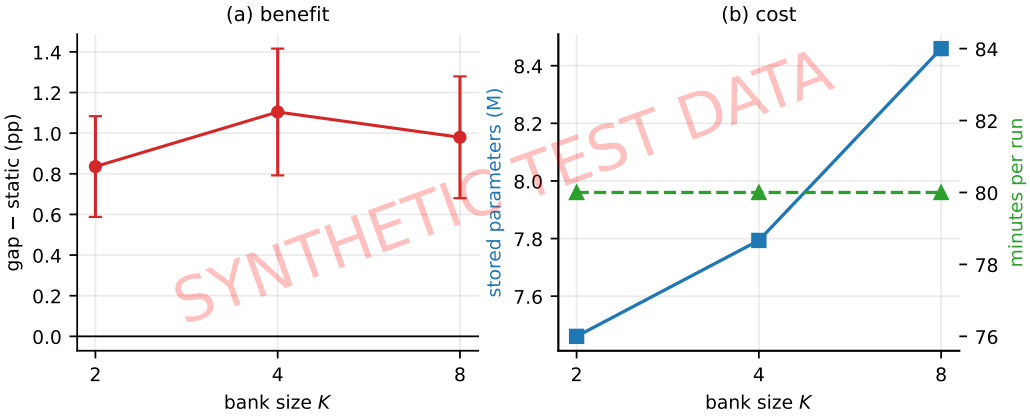


Figure 4: CIFAR-100 width 32, `gap` with $K = 2, 4, 8$. (a) Difference against `static`, 95 % CI. (b) Stored parameters and minutes per run.

$K = 8$ against $K = 2$: +0.144 pp (equivalent) for 1.13× the parameters (Tables 4, 5).

Table 4: Bank size at CIFAR-100 width 32. Differences are against `static`.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	64.733	0.149	+0.836	[+0.588, +1.084]	0.0003	7.46	80.0
4	64.971	0.233	+1.104	[+0.792, +1.416]	0.0006	7.79	80.0
8	64.876	0.232	+0.980	[+0.680, +1.279]	0.0004	8.46	80.0
`static`	63.897	0.117	–	–	–	7.09	20.0

Table 5: Pairwise comparisons between bank sizes.

comparison	Δ (pp)	95 % CI	p	p_{TOST}	verdict
$K = 4$ vs $K = 2$	+0.220	[-0.216, +0.656]	0.2346	0.0744	inconclusive (n=5)
$K = 8$ vs $K = 4$	-0.045	[-0.166, +0.076]	0.3561	0.0002	equivalent (n=5)
$K = 8$ vs $K = 2$	+0.144	[-0.179, +0.467]	0.3035	0.0182	equivalent

6 Tiny-ImageNet

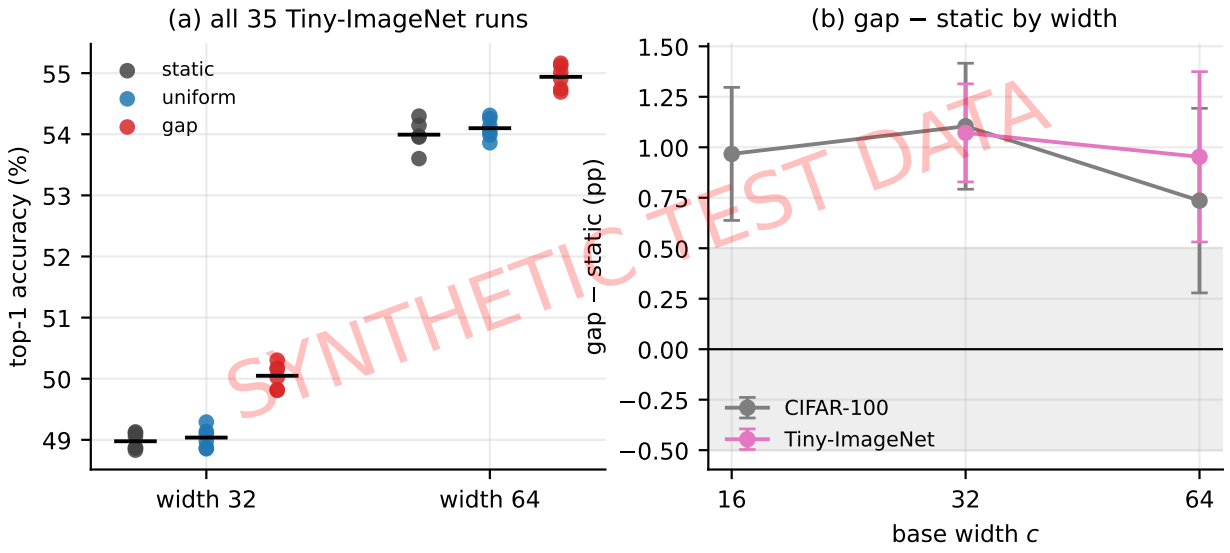


Figure 5: Tiny-ImageNet. (a) Every run at widths 32 and 64. (b) `gap−static` against width on both datasets, 95 % CI; the band is the margin.

`gap−static` is +1.071 pp at width 32 (higher) and +0.953 pp at width 64 (higher (n=5)).

Across the five conditions the Spearman correlation between the static arm's final training loss and `gap−static` is $\rho = +0.60$ (Table 6, Figure 6).

Table 6: Five conditions ordered by the final training loss of the static arm.

dataset	width	static train loss	`gap − static` (pp)	95 % CI	verdict
CIFAR-100	64	0.0244	+0.736	[+0.279, +1.193]	higher
Tiny-ImageNet	64	0.0306	+0.953	[+0.531, +1.375]	higher (n=5)
CIFAR-100	32	0.0619	+1.104	[+0.792, +1.416]	higher (n=5)
Tiny-ImageNet	32	0.0869	+1.071	[+0.828, +1.314]	higher
CIFAR-100	16	0.1119	+0.967	[+0.638, +1.297]	higher

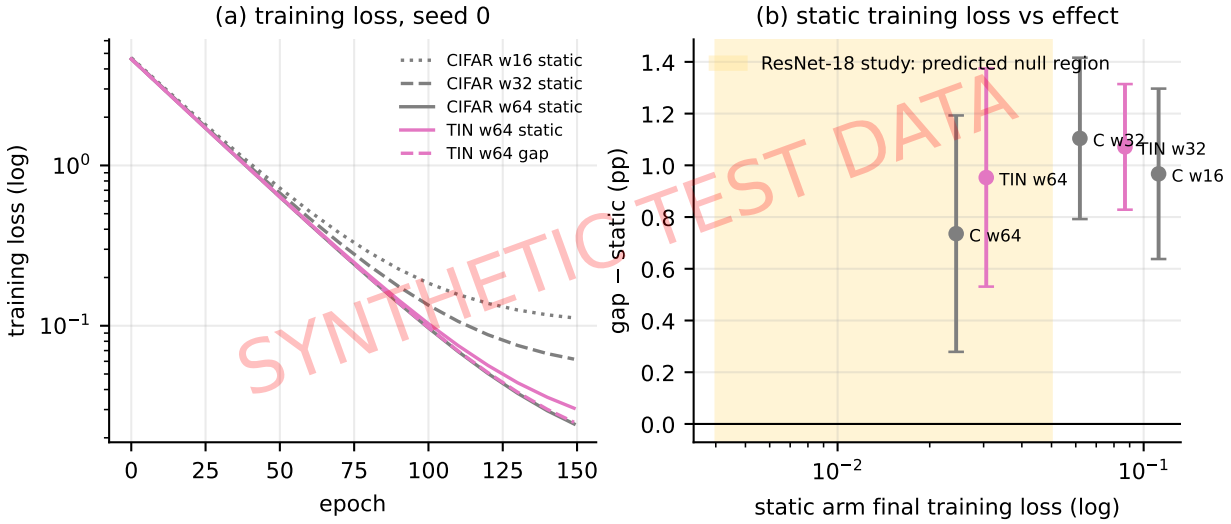


Figure 6: (a) Training loss, seed 0, sampled every ten epochs. (b) `gap−static` against the static arm's final training loss; the shaded region is where the ResNet-18 study predicted a null.

7 Equal parameter budgets

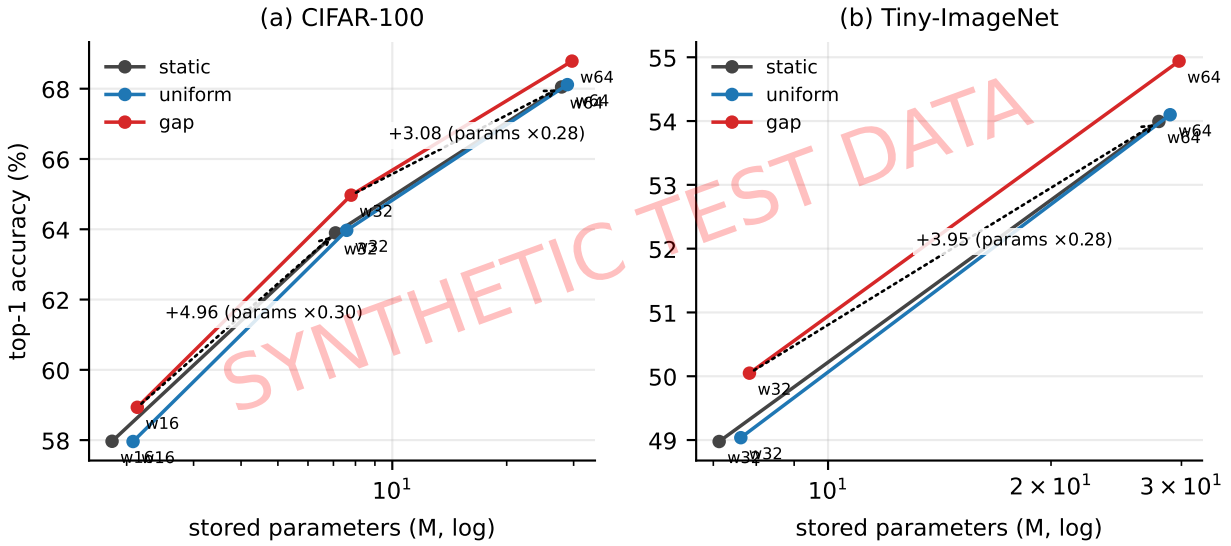


Figure 7: Accuracy against stored parameters. Arrows join `gap` at width c to `static` at width $2c$; the label is the accuracy difference, and the parameter ratio where it is outside $\pm 10\,\%$.

The ResNet-18 comparison of `gap` at width c with `static` at width $2c$ was a matched-budget comparison because a $K = 4$ bank and a doubling of width both multiply the parameters of the dynamic layers by four. That holds only where almost every parameter is in a dynamic layer. Here 0 of 3 pairs are within 10 % and are called matched: CIFAR-100 `gap` w16 vs `static` w32: +4.963 pp, parameter ratio 0.30 (not matched); CIFAR-100 `gap` w32 vs `static` w64: +3.212 pp, parameter ratio 0.28 (not matched); Tiny-ImageNet `gap` w32 vs `static` w64: +3.897 pp, parameter ratio 0.28 (not matched) (Table 7). Table 8 adds the comparison at equal parameters: the static accuracy curve interpolated linearly in log-parameters between the static widths that were run, evaluated at the `gap` network's count. It is an interpolation, not a run, and has no confidence interval.

Table 7: `gap` at width c against `static` at width $2c$. Δ is plain minus routed. The last column is the routed network's parameters divided by the plain one's; a pair is called matched only within 10 %.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	min/run	param ratio
CIFAR-100	`gap` w16 (2.13M)	`static` w32 (7.09M)	58.934	63.897	+4.963	[+4.692, +5.233]	$< 10^{-4}$	40.0 / 20.0	0.30 (not matched)
CIFAR-100	`gap` w32 (7.79M)	`static` w64 (27.89M)	64.971	68.049	+3.212	[+2.959, +3.465]	$< 10^{-4}$	80.0 / 40.0	0.28 (not matched)
Tiny-ImageNet	`gap` w32 (7.83M)	`static` w64 (27.97M)	50.048	53.993	+3.897	[+3.354, +4.441]	$< 10^{-4}$	80.0 / 40.0	0.28 (not matched)

Table 8: `gap` against the static accuracy curve at equal parameters (interpolated, no CI).

dataset	routed	params (M)	routed acc.	static at equal params	Δ (pp)
CIFAR-100	`gap` w16	2.13	58.934	58.637	-0.297
CIFAR-100	`gap` w32	7.79	64.971	64.184	-0.787
Tiny-ImageNet	`gap` w32	7.83	50.048	49.322	-0.726

8 Temperature schedule

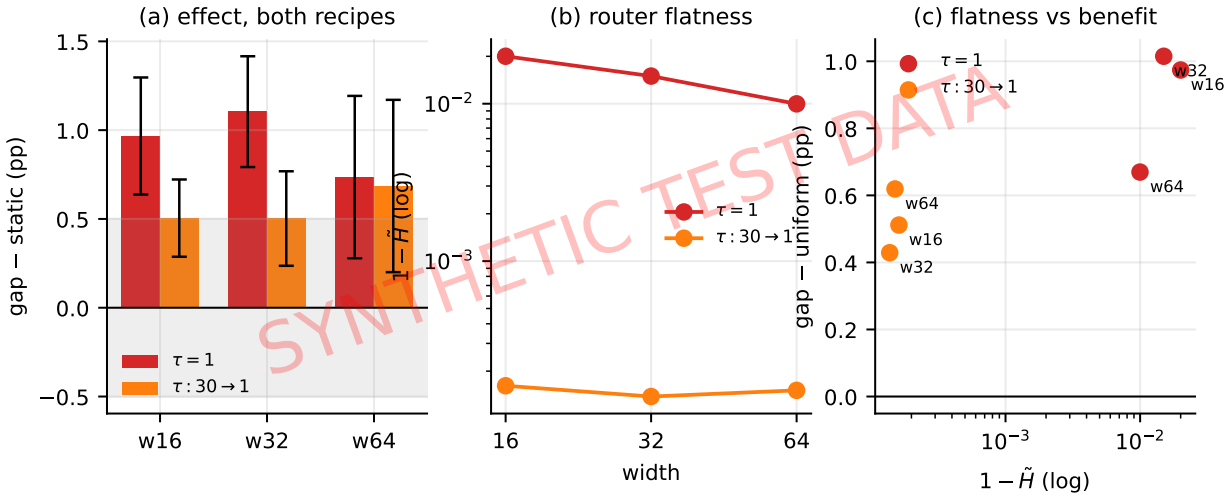


Figure 8: Constant $\tau = 1$ against τ annealed from 30 to 1 over ten epochs, CIFAR-100. (a) `gap−static`. (b) Router flatness. (c) Flatness against benefit over `uniform`.

Annealed minus constant, `gap`: width 16 -0.463 pp (lower); width 32 -0.558 pp (lower (n=5)); width 64 -0.050 pp (equivalent) (Table 9).

Table 9: Temperature schedule, CIFAR-100.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p_{TOST}	Δ	verdict
16	58.934	58.472	-0.463	0.0012	+0.967	+0.505	0.5209	+0.511	higher
32	64.971	64.399	-0.558	0.0059	+1.104	+0.502	0.5086	+0.429	higher
64	68.785	68.735	-0.050	0.6872	+0.736	+0.686	0.8146	+0.619	higher

9 Routing entropy and routing benefit

At width 16 the constant-temperature router's $1 - \tilde{H}$ is 124.0 times the annealed router's; the benefit over `uniform` is +0.974 pp constant and +0.511 pp annealed. Across the 6 width–recipe points the Spearman correlation between $1 - \tilde{H}$ and `gap−uniform` is $\rho = +0.89$ (Table 10).

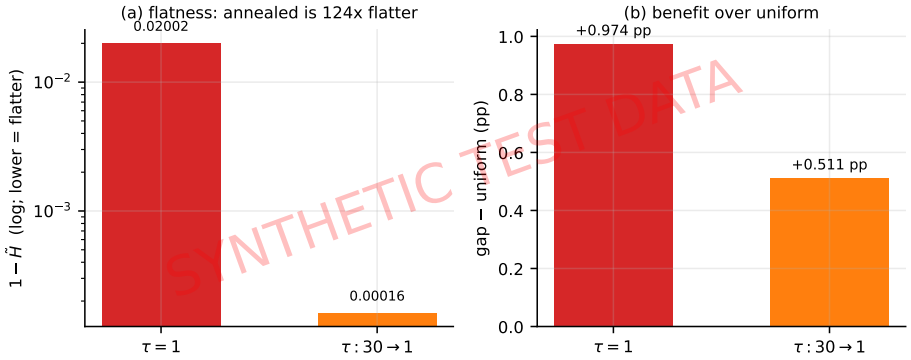


Figure 9: CIFAR-100 width 16, both temperature recipes. (a) Router flatness. (b) Benefit of `gap` over `uniform`.

Table 10: Routing entropy against the benefit over `uniform`.

width	τ	$\tilde{H}$	$1-\tilde{H}$	RMS dev	`gap`−`uniform` (pp)	p
16	1	0.97998	0.02002	0.0589	+0.974	0.0030
16	30→1	0.99984	0.00016	0.0053	+0.511	0.0224
32	1	0.98502	0.01498	0.0510	+1.015	0.0012
32	30→1	0.99986	0.00014	0.0049	+0.429	0.0015
64	1	0.99002	0.00998	0.0416	+0.669	0.0051
64	30→1	0.99985	0.00015	0.0051	+0.619	0.0172

10 Pre-registered predictions

Recorded in `PREREGISTRATION_MODELS.md` on 30 September 2026, before any run of this backbone.

Table 11: Pre-registered predictions and their outcomes.

	prediction	measured	outcome
P1	bank term equivalent to zero in each of 5 conditions	4 of 5	failed
P2	static at width $2c$ beats `gap` at width c in each of 3 pairs	3 of 3	held
P3	routing term positive, $p < 0.05$, in at least 3 of 5 conditions	5 of 5	held

11 Complete results

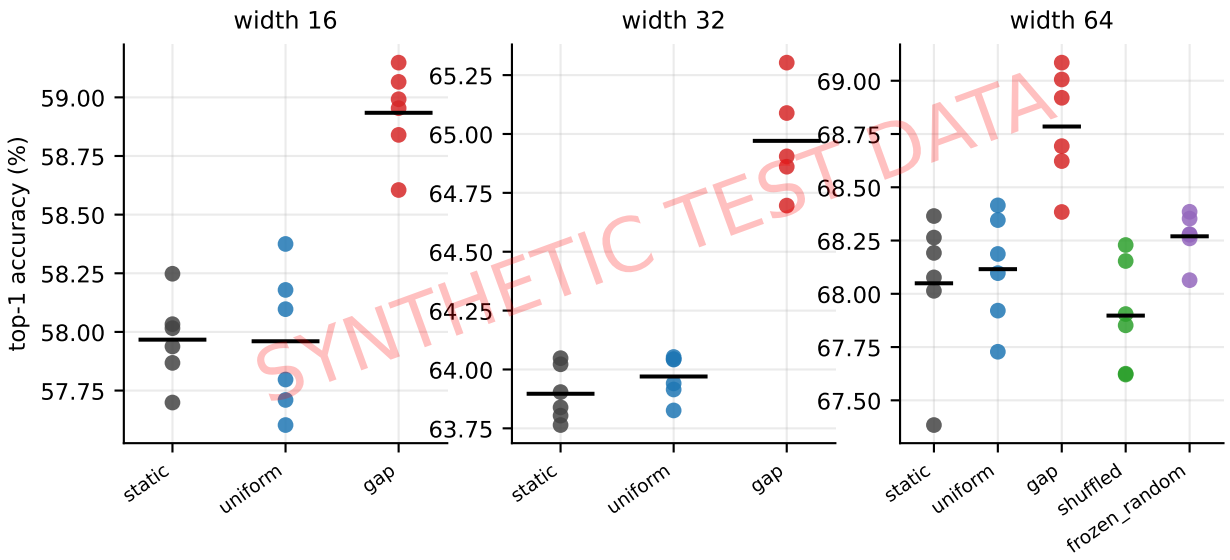


Figure 10: Every CIFAR-100 run at $K = 4$, $\tau = 1$. Bars are arm means.

Table 12: Every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%). – marks a missing or diverged run (Table 1).

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	57.70	58.02	58.25	57.94	58.03	57.87	57.967	0.183
CIFAR-100	16	`uniform`	57.80	58.18	58.38	58.10	57.60	57.71	57.960	0.302
CIFAR-100	16	`gap`	59.15	58.95	58.84	58.60	59.07	58.99	58.934	0.192
CIFAR-100	32	`static`	63.90	64.05	63.76	64.02	63.80	63.84	63.897	0.117
CIFAR-100	32	`uniform`	63.83	64.04	64.05	64.04	63.91	63.94	63.970	0.092
CIFAR-100	32	`gap`	65.30	–	65.09	64.86	64.70	64.91	64.971	0.233
CIFAR-100	64	`static`	68.36	67.38	68.19	68.01	68.26	68.08	68.049	0.350
CIFAR-100	64	`uniform`	68.42	68.35	68.10	67.92	67.73	68.19	68.116	0.259
CIFAR-100	64	`gap`	68.69	68.62	68.92	69.08	68.38	69.01	68.785	0.266
CIFAR-100	64	`shuffled`	68.23	67.62	67.85	68.15	67.91	67.62	67.898	0.256
CIFAR-100	64	`frozen_random`	68.39	68.28	68.06	68.28	68.35	68.26	68.270	0.112
Tiny-ImageNet	32	`static`	48.87	48.87	48.83	49.06	49.11	49.13	48.977	0.136
Tiny-ImageNet	32	`uniform`	49.29	48.85	49.14	49.09	48.98	48.87	49.037	0.170
Tiny-ImageNet	32	`gap`	50.17	49.82	50.02	50.30	50.16	49.81	50.048	0.201
Tiny-ImageNet	64	`static`	53.97	54.30	54.15	53.60	53.95	–	53.993	0.260
Tiny-ImageNet	64	`uniform`	54.05	54.13	53.86	54.31	54.26	53.99	54.099	0.168
Tiny-ImageNet	64	`gap`	55.12	54.69	55.01	54.74	55.16	54.91	54.940	0.195

Table 13: Paired comparisons against `static`. p_W is the Wilcoxon signed-rank p (floor 0.031 at $n = 6$).

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	-0.007	[-0.258, +0.245]	0.9470	0.8438	-0.03	0.0020	equivalent
CIFAR-100	16	`gap`	+0.967	[+0.638, +1.297]	0.0006	0.0312	+3.08	0.9926	higher
CIFAR-100	32	`uniform`	+0.073	[-0.060, +0.207]	0.2162	0.2188	+0.58	0.0002	equivalent
CIFAR-100	32	`gap`	+1.104	[+0.792, +1.416]	0.0006	0.0625	+4.40	0.9971	higher (n=5)
CIFAR-100	64	`uniform`	+0.066	[-0.452, +0.584]	0.7554	1.0000	+0.13	0.0420	equivalent
CIFAR-100	64	`gap`	+0.736	[+0.279, +1.193]	0.0090	0.0312	+1.69	0.8790	higher
CIFAR-100	64	`shuffled`	-0.152	[-0.452, +0.149]	0.2514	0.3125	-0.53	0.0154	equivalent
CIFAR-100	64	`frozen_random`	+0.221	[-0.155, +0.596]	0.1911	0.1562	+0.62	0.0568	inconclusive
Tiny-ImageNet	32	`uniform`	+0.060	[-0.213, +0.333]	0.5953	0.6875	+0.23	0.0045	equivalent
Tiny-ImageNet	32	`gap`	+1.071	[+0.828, +1.314]	$< 10^{-4}$	0.0312	+4.63	0.9991	higher
Tiny-ImageNet	64	`uniform`	+0.127	[-0.364, +0.618]	0.5117	0.6250	+0.32	0.0513	inconclusive (n=5)
Tiny-ImageNet	64	`gap`	+0.953	[+0.531, +1.375]	0.0033	0.0625	+2.80	0.9796	higher (n=5)

Table 14: Configuration and cost. Router parameters in thousands; training loss is the final-epoch mean over seeds; minutes are logged wall clock.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	1.83	0.0	0.1119	10.0
CIFAR-100	16	`uniform`	2.08	0.0	0.1119	40.0
CIFAR-100	16	`gap`	2.13	54.2	0.0819	40.0
CIFAR-100	32	`static`	7.09	0.0	0.0619	20.0
CIFAR-100	32	`uniform`	7.59	0.0	0.0619	80.0
CIFAR-100	32	`gap`	7.79	207.0	0.0469	80.0
CIFAR-100	64	`static`	27.89	0.0	0.0244	40.0
CIFAR-100	64	`uniform`	28.89	0.0	0.0244	160.0
CIFAR-100	64	`gap`	29.70	817.1	0.0206	160.0
CIFAR-100	64	`shuffled`	29.70	817.1	0.0244	160.0
CIFAR-100	64	`frozen_random`	28.89	0.0	0.0244	160.0
Tiny-ImageNet	32	`static`	7.13	0.0	0.0869	20.0
Tiny-ImageNet	32	`uniform`	7.62	0.0	0.0869	80.0
Tiny-ImageNet	32	`gap`	7.83	207.0	0.0644	80.0
Tiny-ImageNet	64	`static`	27.97	0.0	0.0306	40.0
Tiny-ImageNet	64	`uniform`	28.96	0.0	0.0306	160.0
Tiny-ImageNet	64	`gap`	29.78	817.1	0.0250	160.0

Table 15: Every run of this backbone. GPU-hours from logged wall clock.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	9.0
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	17	16.7
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	68.0
CIFAR-100	32	2	constant	`gap`	6	8.0
CIFAR-100	32	8	constant	`gap`	6	8.0
CIFAR-100	16	4	annealed	`gap`	6	4.0
CIFAR-100	32	4	annealed	`gap`	6	8.0
CIFAR-100	64	4	annealed	`gap`	6	16.0
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	18.0
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	17	35.3
total					130	191.0